

Syllabus:- Indefinite Integration
SECTION-I
Straight Objective Type

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If $I_1 = \int \frac{e^x dx}{(1+x^2)^2}$, $I_2 = \int \frac{xe^x dx}{(1+x^2)^2}$ and $I_3 = \int \frac{x^2 e^x dx}{(1+x^2)^2}$ then $I_1 - 2I_2 + I_3 =$

(1) $\frac{e^x}{1+x^2} + C$ (2) $\frac{e^x}{(1+x^2)^2} + C$ (3) $\frac{2e^x}{1+x^2} + C$ (4) $e^x +$
- If $I_1 = \int e^{2x} \sin\left(\frac{\pi}{3} - x\right) \cos x dx$ and $I_2 = \int e^{2x} \cos\left(\frac{\pi}{3} - x\right) \sin x dx$ then $I_1 + I_2 =$

(1) $\frac{\sqrt{3}}{2} e^{2x} \sin x + C$ (2) $\frac{e^{2x} \cos x}{2} + C$

(3) $\frac{\sqrt{3}}{4} e^{2x} + C$ (4) $\frac{e^{2x}}{4} + C$
- $\int \frac{dx}{1 + \sin x + \cos x} =$

(1) $\log\left|1 + \tan \frac{x}{2}\right| + C$ (2) $\log\left|1 - \tan \frac{x}{2}\right| + C$ (3) $\log\left|2 + \tan \frac{x}{2}\right| + C$ (4) $\log\left|1 + 2 \tan \frac{x}{2}\right| + C$
- If $\int \frac{1 - 5 \sin^2 x}{\cos^5 x \sin^2 x} dx = \frac{f(x)}{\cos^5 x} + C$ where C is a constant of integration then f(x) is :

(1) $-\cot x$ (2) $-\operatorname{cosec} x$ (3) $\operatorname{cosec} x$ (4) $\cot x$
- If $\int \frac{dx}{x^3(1+x^6)^{2/3}} = f(x)(1+x^6)^{1/3} + C$ where C is a constant of integration, then f(x) is equal to

(1) $-\frac{1}{2}$ (2) $-\frac{1}{6}$ (3) $-\frac{6}{x}$ (4) $-\frac{x}{2}$
- The integral $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$ is equal to

(1) $\frac{1}{\sqrt{3}} \cot^{-1} \left[\frac{x\sqrt{3}}{\sqrt{x+1}} \right] + C$ (2) $\frac{1}{\sqrt{3}} \tan^{-1} \left[\frac{x}{\sqrt{3(x+1)}} \right] + C$

(3) $\frac{2}{\sqrt{3}} \tan^{-1} \left[\frac{x}{\sqrt{3(x+1)}} \right] + C$ (4) $\frac{2}{\sqrt{3}} \cot^{-1} \left[\frac{x}{\sqrt{x+1}} \right] + C$

7. If $\int \tan^4 x dx = a \tan^3 x + b \tan x + x + C$ where C is a constant of integration then $a + b =$
- (1) $-\frac{2}{3}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) 0
8. If $\int \frac{5 \tan x dx}{\tan x - 2} = x + a \ln |\sin x - 2 \cos x| + C$ where C is a constant of integration then $a =$
- (1) 2.5 (2) 2 (3) 4 (4) 3
9. $\int \frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}} dx =$
- (1) $\frac{2}{3}(1-x)^{3/2} + C$ (2) $\frac{3}{2}(1+x)^{3/2} + C$
- (3) $\frac{3}{2}(1-x)^{3/2} + C$ (4) $\frac{2}{3}(1+x)^{3/2} + C$
10. $\int x^{x^2+1} (2 \ln x + 1) dx =$
- (1) $x^{2x} + C$ (2) $x^2 \ln x + C$ (3) $x^{(x^x)} + C$ (4) $(x^x)^x + C$
11. $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx =$
- (1) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$ (2) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$
- (3) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + C$ (4) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$
12. $\int \sin^5 x dx =$
- (1) $-\cos x + \frac{2}{3} \cos^3 x - \frac{\cos^5 x}{5} + C$ (2) $\cos x + \frac{2}{3} \cos^3 x - \frac{\cos^5 x}{5} + C$
- (3) $\cos x - \frac{2}{3} \cos^3 x - \frac{\cos^5 x}{5} + C$ (4) $\cos x - \frac{2}{3} \cos^3 x + C$
13. The integral $\int x \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) dx (x > 0)$ is equal to:
- (1) $-x + (1+x^2) \tan^{-1} x + C$ (2) $x + (1+x^2) \cot^{-1} x + C$
- (3) $-x + (1+x^2) \cot^{-1} x + C$ (4) $x - (1+x^2) \tan^{-1} x + C$
14. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$ is equal to:
- (1) $\frac{1}{(1+\cot^3 x)} + C$ (2) $-\frac{1}{3(1+\tan^3 x)} + C$ (3) $\frac{\sin^3 x}{(1+\cos^3 x)} + C$ (4) $\frac{\cos^3 x}{3(1+\sin^3 x)} + C$
15. If m is a non-zero number then $\int \frac{x^{5m-1} + 2x^{4m-1}}{(x^{2m} + x^m + 1)^3} dx =$
- (1) $\frac{x^{5m}}{2m(x^{2m} + x^m + 1)^2} + C$ (2) $\frac{x^{4m}}{2m(x^{2m} + x^m + 1)^2} + C$
- (3) $\frac{2m(x^{5m} + x^{4m})}{(x^{2m} + x^m + 1)^2} + C$ (4) $\frac{(x^{5m} - x^{4m})}{2m(x^{2m} + x^m + 1)^2} + C$

16. If $f\left(\frac{x-4}{x+2}\right) = 2x + 1$, ($x \in \mathbb{R} - \{-2\}$), then $\int f(x)dx$ is equal to : (where C is a constant of integration)
- (1) $12 \log_e |1-x| - 3x + C$ (2) $-12 \log_e |1-x| - 3x + C$
 (3) $12 \log_e |1-x| + 3x + C$ (4) $-12 \log_e |1-x| + 3x + C$
17. If $\int \frac{\tan x}{1 + \tan x + \tan^2 x} dx = x - \frac{K}{\sqrt{A}} \tan^{-1} \left(\frac{K \tan x + 1}{\sqrt{A}} \right) + C$ (C is a constant of integration), then the ordered pair (K, A) is equal to
- (1) (2, 1) (2) (2, 3) (3) (-2, 1) (4) (-2, 3)
18. If $I = \int \frac{1}{x} \sqrt{\frac{1-x}{1+x}} dx$, then I equals:
- (1) $\log |x| + \log \left| 1 + \sqrt{1-x^2} \right| + \sin^{-1} \sqrt{x} + C$ (2) $\log |x| - \log \left| 1 - \sqrt{1-x^2} \right| + \tan^{-1} x + C$
 (3) $\log |x| - \log \left| 1 + \sqrt{1-x^2} \right| - \sin^{-1} x + C$ (4) $\log \left| 1 + \sqrt{1-x^2} \right| - \log |x| + \cos^{-1} x + C$
19. The value of $\int \frac{1}{x^2(x^4+1)^{3/4}} dx$ is equal to
- (1) $\left(1 + \frac{1}{x^4}\right)^{1/4} + C$ (2) $(x^4 + 1)^{1/4} + C$ (3) $\left(1 - \frac{1}{x^4}\right)^{1/4} + C$ (4) $-\left(1 + \frac{1}{x^4}\right)^{1/4} + C$
20. If $I = \int \frac{\sqrt{x^2+1}}{x^4} dx$, then I equals
- (1) $-\frac{1}{3} \frac{(x^2+1)^{3/2}}{x^3} + C$ (2) $x^3 (x^2+1)^{-1/2} + C$
 (3) $\frac{\sqrt{x^2+1}}{x^2} + C$ (4) $-\frac{1}{3} \frac{(x^2+1)^{3/2}}{x^2} + C$

SECTION-II

(Integer Type Questions)

This section contains **Five (5)** questions. The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto one digit

21. If $\int \frac{(x-1)(x-2)(x-3)}{(x-4)(x-5)(x-6)} dx = a_1x + a_2 \log|x-4| + a_3 \log|x-5| + a_4 \log|x-6| + C$ then $a_1 + a_2 + a_3 + a_4 =$ (where C is a constant of integration)
22. If $\int \frac{\cos 6x + 6 \cos 4x + 15 \cos 2x + 10}{10 \cos^2 x + 5 \cos x \cos 3x + \cos x \cos 5x} dx = f(x) + C$ where C is a constant of integration then $f(10)$
23. $\int \frac{x^2}{x^4 + 3x^2 + 9} dx = A \tan^{-1} \left(\frac{x^2 - 3}{3x} \right) + \frac{B}{\sqrt{3}} \ln \left| \frac{x^2 - \sqrt{3}x + 3}{x^2 + \sqrt{3}x + 3} \right| + C$ then find the value of $12(A + B)$
24. $\int \frac{e^x(2-x^2)dx}{(1-x)\sqrt{1-x^2}} = k.e^x \left(\frac{1+x}{\sqrt{1-x^2}} \right) + c.$
 Find value of k.

25. $\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}} = k \left(\frac{x-1}{x+2} \right)^{1/4} + c$

Find number of divisors of $30k$.

ANSWER KEY_DPP-15

1.	(1)	2.	(3)	3.	(1)	4.	(1)	5.	(1)	6.	(3)	7.	(1)
8.	(2)	9.	(4)	10.	(4)	11.	(4)	12.	(1)	13.	(1)	14.	(2)
15.	(2)	16.	(2)	17.	(2)	18.	(3)	19.	(4)	20.	(1)	21.	(10)
22.	(20)	23.	(5)	24.	(1)	25.	(8)						